RAYLEIGH QUOTIENTS OF HOLOMORPHIC COCHAINS

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Abstract. Using Rayleigh quotients associated with the combinatorial Hodge star operator, we show that the Rayleigh quotients of holomorphic 1-cochains converge to 1 at a linear rate as the mesh size tends to zero. As applications, we obtain convergence rate estimates for holomorphic 1-cochains and combinatorial period matrices.

1. Introduction

In [8, 9], Wilson [8, 9] applied combinatorial Hodge theory, as developed in [1, 2], to triangulated closed Riemann surfaces by extending it to the complex setting. Wilson then defined the space of holomorphic 1-cochains $\mathcal{H}C^{1,0}(K)$ for a triangulation K and established several of their properties. In particular, for a canonical homology basis Σ, there exists a unique basis $\{\sigma_1, \cdots, \sigma_g\}$ of $\mathcal{H}C^{1,0}(K)$, which we call the canonical basis. Using the canonical basis, Wilson defined the combinatorial period matrix, which is uniquely determined by a canonical homology basis and a triangulation, in analogy with the (conformal) period matrix. In [12], we defined a real-valued vector $\Phi_K = (\varphi_1, \cdots, \varphi_g)$ whose components satisfy

$$\langle \bigstar \sigma_j, \sigma_j \rangle = \langle -i\varphi_j \sigma_j, \sigma_j \rangle,$$

where $\bigstar$ is the skew-adjoint operator on the simplicial cochains of K introduced by Wilson. Then, since $i\bigstar$ is self-adjoint,We define the Rayleigh quotient of σ_j with respect to $i\bigstar$ by

$$\rho_{i\bigstar}(\sigma_j) = \frac{\langle i\bigstar \sigma_j, \sigma_j \rangle}{\langle \sigma_j, \sigma_j \rangle}.$$

By the definition of φ_j, we then have

$$\rho_{i\bigstar}(\sigma_j) = \varphi_j.$$

In this paper, we focus on these Rayleigh quotients of holomorphic 1-cochains and study the differences between holomorphic 1-forms and holomorphic 1-cochains in terms of their Rayleigh quotients. On the other hand, for the Hodge star operator $\star$ on differential forms, the Rayleigh quotient of any holomorphic 1-form ω with respect to $i\star$ is 1. As the main result of this paper, we show that the Rayleigh quotients of holomorphic 1-cochains converge to 1 at a linear rate as the mesh size h tends to zero. In addition, using the main result, we also estimate the convergence rates of holomorphic 1-cochains and combinatorial period matrices.

2. Combinatorial Hodge theory

Let M be a closed smooth Riemannian n-manifold, and let $\Omega^j(M)$ be the space of smooth j-forms on M with the exterior derivative d and the Hodge star operator

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$\star$. The Riemannian metric of M gives an inner product $\langle , \rangle_\Omega$ on $\Omega(M)$. We define $d^* := (-1)^{j(j+1-n)} \star d\star$ and the space $\mathcal{H}\Omega^j(M)$ of harmonic j-forms on M by

$$\mathcal{H}\Omega^j(M) = \left\{\omega \in \Omega^j(M) | d\omega = d^*\omega = 0\right\},$$

and have the Hodge decomposition

$$\Omega^j(M) = d\Omega^{j-1}(M) \oplus \mathcal{H}\Omega^j(M) \oplus d^*\Omega^{j+1}(M).$$

Let K be a C^∞ triangulation of M. Now we identify $|K|$ and M and fix an ordering of the vertices of K. Also, we assume that all simplices in K have nice shape. This means that the shapes do not too thin. Then we denote the i-th vertex of K by p_i and the barycentric coordinate corresponding to p_i by μ_i. Let $C^j(K)$ be the simplicial j-cochains of K with values in $\mathbb{R}$. Given the ordering of vertices, we have a coboundary operator $\delta : C^j(K) \to C^{j+1}(K)$. Since K is a finite complex, we can identify chains and cochains and then for $\sigma \in C^j(K)$, we may write

$$\sigma = \sum_\tau c_\tau \cdot \tau,$$

where $c_\tau \in \mathbb{R}$ and the sum is taken over all j-simplices of K. We write $\tau = [p_0, p_1, \cdots, p_j]$ of K with the vertices in an increasing sequence with respect to the ordering of vertices in K. To construct structures on the cochains $C(K) = \oplus_j C^j(K)$, we assume that the cochains $C(K)$ are equipped with a non-degenerate inner product $\langle , \rangle_C$ such that $C^j(K) \perp C^k(K)$ for $j \neq k$. Then we define the adjoint operator of δ:

Definition 2.1. *The adjoint operator $\delta^* : C^j(K) \to C^{j-1}(K)$ of δ is defined by $\langle \delta^*\sigma_1, \sigma_2\rangle_C = \langle \sigma_1, \delta\sigma_2\rangle_C$.*

Then two operators δ and δ^* give rise to the harmonic cochains as follows.

Definition 2.2. *We define the space $\mathcal{H}C^j(K)$ of harmonic j-cochains of K by*

$$\mathcal{H}C^j(K) = \left\{\sigma \in C^j(K) | \delta\sigma = \delta^*\sigma = 0\right\}.$$

Eckmann showed that an inner product $\langle , \rangle_C$ provides the Hodge decomposition of cochains.

Theorem 2.3 ([4])**.** *There is an orthogonal direct sum decomposition*

$$C^j(K) = \delta C^{j-1}(K) \oplus \mathcal{H}C^j(K) \oplus \delta^* C^{j+1}(K)$$

and $\mathcal{H}C^j(K) \cong H^j(K)$, the cohomology of (K, δ) in degree j.

In [1, 2, 8], one can find several results relating Hodge theory to combinatorial Hodge theory. We now define two maps between differential forms and simplicial cochains. First, we define the *Whitney map* W from $C^j(K)$ into $\mathcal{L}^2\Omega^j(M)$, the completion of $\Omega^j(M)$ with respect to the inner product $\langle \cdot, \cdot\rangle_\Omega$.

Definition 2.4. *For a j-simplex $\tau = [p_0, \cdots, p_j]$ of K, we define $W\tau$ by*

$$W\tau = j! \sum_{i=0}^{j} (-1)^i \mu_i d\mu_0 \wedge \cdots \wedge \widehat{d\mu_i} \wedge \cdots \wedge d\mu_j,$$

where $\widehat{}$ over a symbol means deletion. W is defined on $C(K) = \oplus_{j\in\{0,1,2\}} C^j(K)$ by extending linearly.

Remark 2.5. *The barycentric coordinates μ_j are not even of class C^1, but they are of class C^∞ on the interior of any simplex of K. This implies that $d\mu_j$ is defined and $W\tau$ is well-defined. Therefore dW is also well-defined on $\mathcal{L}^2\Omega^j(M)$.*

The Whitney map W has several properties.

Proposition 2.6 ([7]). *The following hold:*
(1) $W\tau = 0$ *on* $M \backslash \overline{St(\tau)}$,
(2) $dW = W\delta$,
where $\overline{St(\tau)}$ *is the closure of the open star* $St(\tau)$.

Next, we define the de Rham map R from differential forms to cochains, which is given by integration.:

Definition 2.7. *For any differential form* ω *and chain* c, *the de Rham map* R *is defined by*

$$R\omega(c) = \int_c \omega.$$

The de Rham map is a chain map:

Lemma 2.8 ([3]). *The following holds:*

$$\delta R = Rd.$$

Note that the de Rham map is also defined on the image of the Whitney map W. We then have the following result, see [1, 2, 7]:

Theorem 2.9. *The following holds:*

$$RW = Id.$$

In general, $WR \neq Id$. However, Dodziuk and Patodi [2] showed the following approximation theorem.

Definition 2.10. *For a triangulation* K, *We define the mesh* $\eta(K)$ *of* K *by*

$$\eta(K) = \sup r(p, q)$$

where r *means the geodesic distance in* M *and the supremum is taken over all pairs of vertices* p, q *of a* 1*-simplex in* K.

Theorem 2.11 ([1]). *There exist a positive constant* C *and a positive integer* m, *independent of* K, *such that*

$$\|\omega - WR\omega\|_\Omega \leq C \cdot \|(Id + \Delta)^m \omega\|_\Omega \cdot \eta(K)$$

for all C^∞ *differential forms* ω *on* M.

Dodziuk and Patodi showed that, for a sufficiently fine triangulation, cochains provide a good approximation to differential forms with respect to the following Whitney inner product:

$$\langle \sigma, \tau \rangle_C = \langle W\sigma, W\tau \rangle_\Omega$$

for $\sigma, \tau \in C(K)$. It was proved in [1] that the Whitney inner product is non-degenerate.

Theorem 2.12. *Let* $\omega \in \Omega^j(M)$ *and* $R\omega \in C^j(K)$ *have Hodge decompositions*

$$\begin{aligned} \omega &= d\omega_1 + \omega_2 + d^*\omega_3 \\ R\omega &= \delta a_1 + a_2 + \delta^* a_3. \end{aligned}$$

Then

$$\begin{aligned} \|d\omega_1 - W\delta a_1\|_\Omega &\leq \lambda \cdot \|(Id + \Delta)^m \omega\|_\Omega \cdot \eta(K) \\ \|\omega_2 - Wa_2\|_\Omega &\leq \lambda \cdot \|(Id + \Delta)^m \omega\|_\Omega \cdot \eta(K) \\ \|d^*\omega_3 - W\delta^* a_3\|_\Omega &\leq \lambda \cdot \|(Id + \Delta)^m \omega\|_\Omega \cdot \eta(K) \end{aligned}$$

where λ *and* m *are independent of* ω *and* K.

In [8], Wilson defined the combinatorial Hodge star operator on cochains. To this end, he used the following cup product on cochains, which was introduced by Whitney [7]:

Definition 2.13. *We define* $\cup : C^j(K) \otimes C^k(K) \to C^{j+k}(K)$ *by*

$$\sigma \cup \tau = R(W\sigma \wedge W\tau),$$

for $\sigma \in C^j(K)$ *and* $\tau \in C^k(K)$.

Using this cup product, Wilson then defined the star operator as follows:

Definition 2.14. *Let* $\langle , \rangle_C$ *be a positive definite hermitian inner product on* $C(K)$ *such that* $C^j(K) \perp C^k(K)$ *for* $j \neq k$. *For* $\sigma \in C^j(K)$, *we define* $\bigstar\sigma \in C^{n-j}(K)$ *by*

$$\langle \bigstar\sigma, \tau \rangle_C = (\sigma \cup \tau)[M],$$

where $[M]$ *denotes the fundamental class of* M.

This star operator $\bigstar$ has several properties:

Lemma 2.15 ([8]). *The following hold:*

(1) $\bigstar\delta = (-1)^{j+1}\delta^*\bigstar$, *i.e.* $\bigstar$ *is a chain map.*
(2) *For* $\sigma \in C^j(K)$ *and* $\tau \in C^{n-j}(K)$, $\langle \bigstar\sigma, \tau \rangle_C = (-1)^{j(n-j)}\langle \sigma, \bigstar\tau \rangle_C$, *i.e.* $\bigstar$ *is (graded) skew-adjoint.*
(3) $\bigstar$ *induces isomorphisms* $\mathcal{H}C^j(K) \to \mathcal{H}C^{n-j}(K)$ *on harmonic cochains.*

Using Theorem 2.11, Wilson showed that $\bigstar$ converges to the Hodge star operator $\star$ on $\Omega(M)$:

Theorem 2.16 ([8]). *There exist a positive constant* C *and a positive integer* m, *independent of* K, *such that*

$$\| \star \omega - W\bigstar R\omega\|_\Omega \leq C \cdot \|(Id + \Delta)^m \omega\|_\Omega \cdot \eta(K),$$

for all C^∞ *differential forms* ω *on* M.

Under the assumption that the cochains $C(K)$ are equipped with the Whitney inner product, Wilson also showed that $\bigstar$ is compatible with the Hodge star operators on $C(K)$ and $\Omega(M)$:

Theorem 2.17 ([8]). *Let* $\omega \in \Omega^j(M)$ *and* $R\omega \in C^j(K)$ *have the Hodge decompositions*

$$\begin{aligned} \omega &= d\omega_1 + \omega_2 + d^*\omega_3 \\ R\omega &= \delta a_1 + a_2 + \delta^* a_3. \end{aligned}$$

Then there exist a positive constant C *and a positive integer* m, *independent of* K, *such that*

$$\begin{aligned} \| \star d\omega_1 - W\bigstar\delta a_1\|_\Omega &\leq C \cdot (\|(Id + \Delta)^m \omega\|_\Omega + \|(Id + \Delta)^m d\omega_1\|_\Omega) \cdot \eta(K), \\ \| \star \omega_2 - W\bigstar a_2\|_\Omega &\leq C \cdot (\|(Id + \Delta)^m \omega\|_\Omega + \|(Id + \Delta)^m \omega_2\|_\Omega) \cdot \eta(K), \\ \| \star d^*\omega_3 - W\bigstar\delta^* a_3\|_\Omega &\leq C \cdot (\|(Id + \Delta)^m \omega\|_\Omega + \|(Id + \Delta)^m d^*\omega_3\|_\Omega) \cdot \eta(K). \end{aligned}$$

3. Holomorphic 1-forms on Riemann surfaces

In this section, we recall the definitions and properties of holomorphic 1-forms and period matrices. Let M be a closed Riemann surface of genus g, $\Omega^j(M)$ be the space of smooth j-forms on M with complex values, and $\Omega(M) = \oplus_{j\in\{0,1,2\}}\Omega^j(M)$.

Using the Hodge star operator $\star$, the space of holomorphic 1-forms on M is defined by

$$\mathcal{H}\Omega^{1,0}(M) = \left\{\omega \in \mathcal{H}\Omega^1(M) \mid \star\omega = -i\omega\right\}.$$

Now we define an inner product $\langle \cdot, \cdot \rangle_\Omega$ on $\Omega(M)$ by

$$\langle \omega_1, \omega_2 \rangle_\Omega = \iint_M \omega_1 \wedge \star\overline{\omega_2}.$$

Next, we take a canonical homology basis $a_1, \cdots, a_g, b_1, \cdots, b_g$ for the first homology group, satisfying the following property: the intersection number of two basis elements is non-zero only for a_j and b_j, in which case it is equal to one. We then define the periods of a closed 1-form ω with respect to this basis by

$$\int_{a_j} \omega, \qquad \int_{b_k} \omega,$$

for $1 \leq j, k \leq g$. These periods satisfy the following relations, which are called Riemann's bi-linear relations: for two closed 1-forms ω_1, ω_2,

$$\langle \star\omega_1, \omega_2 \rangle_\Omega = \sum_{j=1}^{g} \left(\int_{a_j} \omega_1 \int_{b_j} \overline{\omega_2} - \int_{b_j} \omega_1 \int_{a_j} \overline{\omega_2} \right).$$

Especially, for any $\omega_1, \omega_2 \in \mathcal{H}\Omega^{1,0}(M)$,

$$\sum_{j=1}^{g} \left(\int_{a_j} \omega_1 \int_{b_j} \omega_2 - \int_{b_j} \omega_1 \int_{a_j} \omega_2 \right) = 0.$$

This relation implies that if all $\int_{a_j} \omega$ of a holomorphic 1-form ω are zero, then $\omega = 0$. Therefore we have the following isomorphism:

$$\mathcal{H}\Omega^{1,0}(M) \ni \omega \longmapsto \left(\int_{a_1} \omega, \cdots, \int_{a_g} \omega \right) \in \mathbb{C}^g.$$

This implies that holomorphic 1-forms are characterized by their A-periods. Using this property, we define a unique basis $\theta_1, \cdots, \theta_g$ of $\mathcal{H}\Omega^{1,0}(M)$ satisfying $\int_{a_k} \theta_j = \delta_{jk}$ for $1 \leq j, k \leq g$. We call this basis the canonical basis of $\mathcal{H}\Omega^{1,0}(M)$, determined by the canonical homology basis Σ. We then define the period matrix $\Pi = (\pi_{jk})_{1 \leq j,k \leq g}$ by $\pi_{jk} = \int_{b_k} \theta_j$. It is known that period matrices lie in the Siegel upper half space.

4. Combinatorial Hodge theory on Riemann surfaces

In [9], Wilson applied combinatorial Hodge theory to triangulated Riemann surfaces by extending the definitions of combinatorial Hodge theory to the complex setting. Let M be a closed Riemann surface with a triangulation K, and let $C^j(K)$ be the simplicial cochains with values in $\mathbb{C}$. Suppose that the cochains $C(K)$ are equipped with a non-degenerate inner product such that $C^i(K)$ and $C^j(K)$ are orthogonal for $i \neq j$. Then the star operator $\bigstar$ on the cochains $C(K)$ is defined by

$$\langle \bigstar\sigma, \tau \rangle_C = (\sigma \cup \overline{\tau})[M],$$

where $\cup$ is extended $\mathbb{C}$-linearly. For complex cochains, we also have the Hodge decomposition

$$C^1(K) = \delta C^0(K) \oplus \mathcal{H}C^1(K) \oplus \delta^* C^2(K).$$

By Lemma 2.15, we regard $\bigstar$ as a skew-adjoint isomorphism from $\mathcal{H}C^1(K)$ to itself. We then define holomorphic 1-cochains as follows:

Definition 4.1. *Let $\langle , \rangle_C$ be a hermitian inner product on the complex valued simplicial 1-cochains which is $\mathbb{R}$-valued on $\mathbb{R}$-cochains. We define the space $\mathcal{H}C^{1,0}(K)$ of holomorphic 1-cochains to be the span of the eigenvectors for non-positive imaginary eigenvalues of $\bigstar$ and the space $\mathcal{H}C^{0,1}(K)$ of anti-holomorphic 1-cochains to be the span of the eigenvectors for non-negative imaginary eigenvalues of $\bigstar$.*

Now we assume that the cochains $C(K)$ are equipped with the Whitney inner product. Note that the Whitney inner product is $\mathbb{R}$-valued on real cochains. Then, by [9], we have the following properties of $\mathcal{H}C^{1,0}(K)$ and $\mathcal{H}C^{0,1}(K)$.

Lemma 4.2 ([9])**.** *Let M be a closed Riemann surface of genus g with a canonical homology basis Σ and a triangulation K. Then, the following hold:*
(1) $\mathcal{H}C^1(K) = \mathcal{H}C^{1,0}(K) \oplus \mathcal{H}C^{0,1}(K)$.
(2) $\dim_{\mathbb{C}} \mathcal{H}C^{1,0}(K) = \dim_{\mathbb{C}} \mathcal{H}C^{0,1}(K) = g$.
(3) *Complex conjugation maps $\mathcal{H}C^{1,0}(K)$ to $\mathcal{H}C^{0,1}(K)$ and vice versa.*

Next we define *combinatorial periods*.

Definition 4.3. *Let M be a closed Riemann surface of genus g with a canonical homology basis Σ and a triangulation K. We define the combinatorial periods of $\sigma \in \mathcal{H}C^1(K)$ by the following complex numbers:*

$$\sigma(a_j),\ \sigma(b_j) \quad for\ 1 \le j \le g.$$

Combinatorial periods of holomorphic 1-cochains also satisfy Riemann's bi-linear relations.

Theorem 4.4 ([9])**.** *For σ, $\tau \in \mathcal{H}C^{1,0}(K)$, we have*

$$\sum_{j=1}^{g} (\sigma(a_j)\tau(b_j) - \sigma(b_j)\tau(a_j)) = 0.$$

Wilson [9] showed the following relation. For $\sigma, \tau \in \mathcal{H}C^{1,0}(K)$,

$$\langle \bigstar\sigma, \tau\rangle_C = \sum_{j=1}^{g} (\sigma(a_j)\overline{\tau(b_j)} - \sigma(a_j)\overline{\tau(b_j)}),$$

and this yields the properties of the combinatorial periods as follows.

Corollary 4.5 ([9])**.** *Let σ be a holomorphic 1-cochain.*
(1) *If all A-periods $\sigma(a_j)$, $1 \le j \le g$ or all B-periods $\sigma(b_j)$, $1 \le j \le g$ are vanish, then $\sigma = 0$.*
(2) *If all A-periods $\sigma(a_j)$, $1 \le j \le g$ and all B-periods $\sigma(b_j)$, $1 \le j \le g$ are real, then $\sigma = 0$.*

For any basis $\{\tau_1, \cdots, \tau_g\}$ for $\mathcal{H}C^{1,0}(K)$, we consider the following equation of $(c_{ij})_{1\le i,j\le g}$:

$$\sum_{i=1}^{g} c_{ij}\tau_i(a_k) = \delta_{jk}.$$

By Corollary 4.5 (1), the matrix $(c_{ij})_{1\le i,j\le g}$ is uniquely determined by a triple (M, Σ, K). Then we put $\sigma_j = \sum_{j=1}^{g} c_{ij}\tau_i$. This basis $\{\sigma_1, \cdots, \sigma_g\}$ is called the canonical basis of $\mathcal{H}C^{1,0}(K)$.

Using the canonical basis, we define combinatorial period matrices.

Definition 4.6. *Let M be a closed Riemann surface of genus g with a canonical homology basis Σ and a triangulation K. Let $\{\sigma_1, \cdots, \sigma_g\}$ be the canonical basis of $\mathcal{H}C^{1,0}(K)$. Then the combinatorial period matrix $\Pi_K = (\pi^K_{jk})_{1\le j,k\le g}$ of M is defined by $\pi^K_{jk} = \sigma_j(b_k)$.*

Theorem 4.7 ([9])**.** *Combinatorial period matrices are symmetric and their imaginary parts are positive definite.*

This theorem implies that combinatorial period matrices lie in the Siegel upper half space. In addition, Wilson showed that combinatorial period matrices give an approximation of (conformal) period matrices.

Theorem 4.8 ([9])**.** *Let M be a closed Riemann surface with a canonical homology basis Σ, and let Π be the period matrix. Let $\{K_n\}_{n\in\mathbb{N}}$ be a sequence of triangulations of M with mesh converging to zero. Then for each n, we have the combinatorial period matrix Π_{K_n}, given by K_n, and*

$$\lim_{n\to\infty} \Pi_{K_n} = \Pi.$$

To prove this approximation theorem, Wilson showed the following lemmas, which also use for the proof of Theorem 5.3.

Lemma 4.9 ([9])**.** *Let M be a closed Riemann surface with a triangulation K. For any $\omega \in \mathcal{H}\Omega^{1,0}(M)$ which has the following decomposition*

$$R\omega = \delta g + h_1 + h_2 + \delta^* k,$$

where $h_1 \in \mathcal{H}C^{1,0}(K)$ and $h_2 \in \mathcal{H}C^{0,1}(K)$. Then there exist positive constant C, dependent on ω but independent of K, such that

$$\|Wh_1 - \omega\|_\Omega \le C \cdot \eta(K).$$

The original proof of the above lemma in [9] provides the following lemma. We can find the proof in [12]

Lemma 4.10 ([12])**.** *Let M be a closed Riemann surface with a triangulation K. For any $\omega \in \mathcal{H}\Omega^{1,0}(M)$ which has the following decomposition*

$$R\omega = \delta g + h_1 + h_2 + \delta^* k,$$

where $h_1 \in \mathcal{H}C^{1,0}(K)$ and $h_2 \in \mathcal{H}C^{0,1}(K)$. Then there exist positive constant C, dependent on ω but independent of K, such that

$$\|W\bigstar h_1 - \star\omega\|_\Omega \le C \cdot \eta(K).$$

In the proof of Theorem 4.8 (Theorem 7.2 in [9]), for a sequence K_n of triangulations whose mesh tends to zero, Wilson stated that, for the holomorphic part h_j^n of $R^n\theta_j \in C^1(K_n)$, Lemma 4.9 (Lemma 7.1 in [9]) provides

$$\lim_{n\to\infty} h_j^n(a_k) = \int_{a_k} \theta_j = \delta_{jk}. \tag{4.1}$$

However, in [10], Wilson remarked that (4.1) does not follow from the lemma, since the convergence in the lemma is with respect to the $\mathcal{L}^2$-norm, while integration is not a bounded operator on smooth forms with respect to this norm. Wilson also stated that, by applying the following lemma, (4.1) holds because we are considering smooth closed differential forms.

Lemma 4.11 ([10])**.** *Let ω_n be a sequence of smooth closed differential forms on a closed Riemannian manifold that converges in L^2 to a smooth form ω. Then, for any cycle C, the sequence $\int_C \omega_n$ converges to $\int_C \omega$.*

In [11], for a fixed triangulation, we obtain the difference between the period matrix and the combinatorial period matrix of a closed Riemann surface with a canonical homology basis.

Theorem 4.12 ([11]). *Let M be a closed Riemann surface of genus g with a canonical homology basis Σ and a triangulation K, and let Π be the period matrix and Π_K the combinatorial period matrix. Let $\{\theta_1, \cdots, \theta_g\}$ be the canonical basis of $\mathcal{H}\Omega^{1,0}(M)$ and $\{\sigma_1, \cdots, \sigma_g\}$ the canonical basis of $\mathcal{H}C^{1,0}(K)$. Then the following equation holds:*

$$\Pi = \overline{\Pi_K} - \overline{\Lambda_K},$$

where $\Lambda_K = (\langle W\sigma_j, \star\theta_k\rangle_\Omega)_{1\le j,k\le g}$.

Proposition 4.13 ([11]). *For a closed Riemann surface M of genus g with a canonical homology basis Σ and a triangulation K, Λ_K lies in the Siegel upper half space.*

Next we introduce a correspondence between holomorphic 1-forms and holomorphic 1-cochains as follows.

Definition 4.14. *For $\omega \in \mathcal{H}\Omega^{1,0}(M)$, we define $\iota_\omega \in \mathcal{H}C^{1,0}(K)$ which satisfies*

$$\iota_\omega(a_j) = \int_{a_j} \omega,$$

for $1 \le j \le g$.

Lemma 4.15 ([12]). *The map ι is an isomorphism from $\mathcal{H}\Omega^{1,0}(M)$ into $\mathcal{H}C^{1,0}(K)$.*

For this map ι, we show the following theorem which is the main result of this paper.

Theorem 4.16 ([12]). *Let M be a closed Riemann surface of genus g with a canonical homology basis Σ and a triangulation K, and let $\{\theta_1, \cdots, \theta_g\}$ be the canonical basis of $\mathcal{H}\Omega^{1,0}(M)$, $\{\sigma_1, \cdots, \sigma_g\}$ the canonical basis of $\mathcal{H}C^{1,0}(K)$ and $\Pi_K = (\pi^K_{jk})_{1\le j,k\le g}$ the combinatorial period matrix. Then there exists a vector $\Phi_K = (\varphi_1, \cdots, \varphi_g) \in (0,1]^g$ such that*

$$\langle \bigstar\sigma_j, \sigma_j\rangle_C = \langle -i\varphi_j\sigma_j, \sigma_j\rangle_C.$$

In addition, we have

$$\|W\sigma_j - \theta_j\|_\Omega = \sqrt{2\mathrm{Im}\pi^K_{jj}\left(\frac{1}{\varphi_j} - 1\right)},$$

and

$$\|W\iota_\omega - \omega\|_\Omega \le \sum_{j=1}^{g} \left| \int_{a_j} \omega \right| \cdot \sqrt{2\mathrm{Im}\pi^K_{jj}\left(\frac{1}{\varphi_j} - 1\right)},$$

for all $\omega \in \mathcal{H}\Omega^{1,0}(M)$.

Theorem 4.17 ([12]). *Let M be a closed Riemann surface of genus 1 (complex torus) with a canonical homology basis Σ, K a triangulation of M, and $\varphi_1 \in (0,1]$ which satisfies $\langle \bigstar\sigma_1, \sigma_1\rangle_C = \langle -i\varphi_1\sigma_1, \sigma_1\rangle_C$, where $\{\sigma_1\}$ is the canonical basis of $\mathcal{H}C^{1,0}(K)$. Then $\varphi_1 = 1$.*

Theorem 4.18 ([12]). *Let M be a closed Riemann surface of genus g with a canonical homology basis Σ, ω arbitrary holomorphic 1-form on M. In the case of $g = 1$, for any triangulation K of M, we have*

$$\|W\iota_\omega - \omega\|_\Omega = 0.$$

In the case of $g > 1$, for any sequence $\{K_n\}_{n\in\mathbb{N}}$ of triangulations of M with the mesh converging to zero, we have

$$\lim_{n\to\infty} \|W\iota^n_\omega - \omega\|_\Omega = 0,$$

where ι^n is the isomorphism from $\mathcal{H}\Omega^{1,0}(M)$ into $\mathcal{H}C^{1,0}(K_n)$.

5. Main results

Since $\bigstar$ is skew-adjoint, $i\bigstar$ is self-adjoint on $\mathcal{H}C^{1,0}(K)$, i.e.,

$$\langle i\bigstar h_1, h_2\rangle_C = \langle h_1, i\bigstar h_2\rangle_C$$

for any holomorphic 1-cochains h_1 and h_2. Then we can consider the Rayleigh quotients ρ of the canonical basis $\{\sigma_1, \cdots, \sigma_g\}$ with respect to $i\bigstar$. of $\mathcal{H}C^{1,0}(K)$ for the combinatorial Hodge star operator $i\bigstar$:

$$\rho_{i\bigstar}(h) = \frac{\langle i\bigstar h, h\rangle_C}{\langle h, h\rangle_C},$$

for any $h \in \mathcal{H}C^{1,0}(K)$. Especially, we study the behavior of the Rayleigh quotients of the canonical basis $\{\sigma_1, \cdots, \sigma_g\}$ of $\mathcal{H}C^{1,0}(K)$ for $i\bigstar$:

$$\rho_{i\bigstar}(\sigma_j) = \frac{\langle i\bigstar \sigma_j, \sigma_j\rangle_C}{\langle \sigma_j, \sigma_j\rangle_C}.$$

Since $\rho_{i\bigstar}(\sigma_j) = \varphi_j$, by Theorem 4.17 and 4.16, we see that the Rayleigh quotients of the canonical basis of holomorphic 1-cochains satisfy the following properties.

Proposition 5.1. *For the canonical basis $\{\sigma_1, \cdots, \sigma_g\}$, their Rayleigh quotients hold*

$$0 < \rho_{i\bigstar}(\sigma_j) \leq 1$$

for $1 \leq j \leq g$. In addition, the case of $g = 1$, we obtain

$$\rho_{i\bigstar}(\sigma_1) = 1.$$

By considering the Rayleigh quotients of the canonical basis of $\mathcal{H}C^{1,0}(K)$ and applying Theorem 4.16, we see that, for any holomorphic 1-cochain ω, the closeness between $W\iota_\omega$ and ω depends on the Rayleigh quotients.

Theorem 5.2. *Let M be a closed Riemann surface of genus g with a canonical homology basis Σ and a triangulation K, and let $\{\theta_1, \cdots, \theta_g\}$ be the canonical basis of $\mathcal{H}\Omega^{1,0}(M)$, $\{\sigma_1, \cdots, \sigma_g\}$ the canonical basis of $\mathcal{H}C^{1,0}(K)$ and $\Pi_K = (\pi^K_{jk})_{1\leq j,k\leq g}$ the combinatorial period matrix. Then*

$$\|W\iota_\omega - \omega\|_\Omega \leq \sum_{j=1}^{g} \left| \int_{a_j} \omega \right| \cdot \sqrt{2\mathrm{Im}\pi^K_{jj}\Big(\frac{1}{\rho_{i\bigstar}(\sigma_j)} - 1\Big)},$$

with equality when $\omega \in \{\theta_1, \cdots, \theta_g\}$.

It is known that for the Hodge star operator $\star$, the Rayleigh quotient of $i\star$ is

$$\rho_{i\star}(\omega) = 1$$

for all holomorphic 1-forms ω. Therefore, the following theorem shows that the Rayleigh quotients of holomorphic 1-cochains converge to those of holomorphic 1-forms as the mesh tends to zero, at a linear rate.

Theorem 5.3. *Let M be a closed Riemann surface of genus $g > 1$ with a canonical homology basis Σ. For a sequence $\{K_n\}_{n\in\mathbb{N}}$ of triangulations of M with the mesh $\eta(K_n)$ converging to zero,*

$$\big|\rho_{i\bigstar}(\sigma^n_j) - 1\big| = O\big(\eta(K_n)\big), \quad \text{as } n \to \infty,$$

where $\{\sigma^n_1, \cdots, \sigma^n_g\}$ is the canonical basis for $\mathcal{H}C^{1,0}(K_n)$ and $n \in \mathbb{N}$.

Proof. Using the Cauchy-Schwarz inequality, we compute

$$\begin{aligned} 0 \leq \big(1 - \rho_{i\bigstar}(\sigma_j^n)\big)\|\sigma_j^n\|_C^2 &= \langle \sigma_j^n, \sigma_j^n\rangle_C - \rho_{i\bigstar}(\sigma_j^n)\langle \sigma_j^n, \sigma_j^n\rangle_C \\ &= \langle \sigma_j^n, \sigma_j^n\rangle_C - \langle i\bigstar\sigma_j^n, \sigma_j^n\rangle_C \\ &= \langle \sigma_j^n - i\bigstar\sigma_j^n, \sigma_j^n\rangle_C \\ &\leq \|\sigma_j^n - i\bigstar\sigma_j^n\|_C \cdot \|\sigma_j^n\|_C, \end{aligned}$$

and then

$$0 \leq \big(1 - \rho_{i\bigstar}(\sigma_j^n)\big)\|\sigma_j^n\|_C \leq \|\sigma_j^n - i\bigstar\sigma_j^n\|_C. \tag{5.1}$$

Next we consider the upper bound of (5.1). Let $\{\omega_1, \cdots, \omega_g\}$ be an orthogonal basis of $\mathcal{H}\Omega^{1,0}(M)$ and R^n the de Rham map from $\Omega(M)$ to $C(K_n)$. By the Hodge decomposition and $\mathcal{H}C^1(K_n) = \mathcal{H}C^{1,0}(K_n) \oplus \mathcal{H}C^{0,1}(K_n)$, we obtain

$$R^n\omega_j = \delta^* k_j^n + h_j^n + \widetilde{h}_j^n + \delta g_j^n$$

for any $n \in \mathbb{N}$, where $h_j^n \in \mathcal{H}C^{1,0}(K_n)$ and $\widetilde{h}_j^n \in \mathcal{H}C^{0,1}(K_n)$.

First, we show that the number of K_n, such that $\{h_1^n, \cdots, h_g^n\}$ is not a basis of $\mathcal{H}C^{1,0}(K_n)$, is finite. Now we assume that the number is infinite. Then there exist $j \in \{1, \cdots, g\}$ and a subsequence $\{K_m\}$ of $\{K_n\}$ such that each h_j^m is generated by the other elements, i.e.,

$$h_j^m = \sum_{p\neq j} c_{jp}^m h_p^m,$$

for all $m \in \mathbb{N}$, where $c_{jp}^m \in \mathbb{C}$. Since $h_j^m - \sum_{p\neq j} c_{jp}^m h_p^m$ is the holomorphic part of $R^m(\omega_j - \sum_{p\neq j} c_{jp}^m \omega_p)$, by Lemma 4.9, we have

$$\lim_{m\to\infty} \Big\|\omega_j - \sum_{p\neq j} c_{jp}^m\omega_p\Big\|_\Omega = 0. \tag{5.2}$$

Also, since $\{\omega_1, \cdots, \omega_g\}$ is an orthogonal basis,

$$\Big\|\omega_j - \sum_{p\neq j} c_{jp}^m\omega_p\Big\|_\Omega^2 = \|\omega_j\|_\Omega^2 + \sum_{p\neq j} |c_{jp}^m|^2\|\omega_p\|_\Omega^2, \tag{5.3}$$

and therefore

$$0 \leq |c_{jp}^m|^2\|\omega_p\|_\Omega^2 \leq \Big\|\omega_j - \sum_{p\neq j} c_{jp}^m\omega_p\Big\|_\Omega^2.$$

By (5.2), we obtain $\lim_{m\to\infty}|c_{jp}^m| = 0$ and (5.3) implies $\|\omega_j\|_\Omega = 0$. This is a contradiction since $\{\omega_1, \cdots, \omega_g\}$ is a basis.

Here we assume that $\{h_1^n, \cdots, h_g^n\}$ is a basis of $\mathcal{H}C^{1,0}(K_n)$ for all $n \in \mathbb{N}$. Then, for any $n \in \mathbb{N}$, we may write

$$\sigma_j^n = \sum_{p=1}^{g} \widetilde{c}_{j\ell}^n h_\ell^n,$$

for $1 \leq j \leq g$, where $\widetilde{c}_{j\ell}^n \in \mathbb{C}$. Next we consider $\lim_{n\to\infty}\widetilde{c}_{j\ell}^n$. Let $(\widetilde{d}_{\ell j}^n)_{1\leq \ell,j\leq g}$ be the inverse matrix of $(\widetilde{c}_{j\ell}^n)_{1\leq j,\ell\leq g}$. This matrix provides

$$h_\ell^n = \sum_{j=1}^{g} \widetilde{d}_{\ell j}^n \sigma_j^n,$$

and

$$h_\ell^n(a_k) = \sum_{j=1}^{g} \widetilde{d}_{\ell j}^n \sigma_j^n(a_k) = \sum_{j=1}^{g} \widetilde{d}_{\ell j}^n \cdot \delta_{jk} = \widetilde{d}_{\ell k}^n,$$

for $1 \le \ell, k \le g$. By Lemma 4.9 and 4.11, we have

$$\lim_{n\to\infty} \widetilde{d}^n_{\ell k} = \lim_{n\to\infty} h^n_\ell(a_k) = \int_{a_k} \omega_\ell,$$

for $1 \le \ell, k \le g$. Namely the matrix $(\widetilde{d}^n_{\ell j})_{1\le \ell, j\le g}$ converges to $(\int_{a_j} \omega_\ell)_{1\le \ell,j\le g}$, and therefore $(\widetilde{c}^n_{j\ell})_{1\le j,\ell\le g} = (\widetilde{d}^n_{\ell j})^{-1}_{1\le \ell,j\le g}$ also converges to a matrix $(s_{j\ell})_{1\le j,\ell\le g}$, where each $s_{j\ell}$ is determined by $\int_{a_1}\omega_1, \cdots, \int_{a_g}\omega_1, \cdots, \int_{a_1}\omega_g, \cdots, \int_{a_g}\omega_g$. This implies that there exist some positive constant M and natural number N_1 such that

$$|\widetilde{c}^n_{j\ell}| < M$$

for $n \ge N_1$ and $1 \le j, \ell \le g$. Since $\sigma^n_j = \sum_{\ell=1}^g \widetilde{c}^n_{\ell j} h^n_\ell$, we have

$$\begin{aligned}
\|\sigma^n_j - i\bigstar\sigma^n_j\|_C &= \left\| \sum_{\ell=1}^g \widetilde{c}^n_{j\ell}\big(h^n_\ell - i\bigstar h^n_\ell\big) \right\|_C \\
&\le \sum_{\ell=1}^g |\widetilde{c}^n_{j\ell}| \cdot \|Wh^n_\ell - iW\bigstar h^n_\ell\|_\Omega \\
&= \sum_{\ell=1}^g |\widetilde{c}^n_{j\ell}| \cdot \|Wh^n_\ell - \omega_\ell + i\star\omega_\ell - iW\bigstar h^n_\ell\|_\Omega \\
&\le \sum_{\ell=1}^g |\widetilde{c}^n_{j\ell}| \cdot \Bigg(\|Wh^n_\ell - \omega_\ell\|_\Omega + \|\star\omega_\ell - W\bigstar h^n_\ell\|_\Omega \Bigg).
\end{aligned}$$

Note that any holomorphic 1-form ω satisfies $\omega - i\star\omega = 0$. By Lemma 4.9 and 4.10, there exist positive constants C_ℓ, independent of $\{K_n\}$, such that

$$\|Wh^n_\ell - \omega_\ell\|_\Omega + \|W\bigstar h^n_\ell - \star\omega_\ell\|_\Omega \le C_\ell \cdot \eta(K_n).$$

Thus we have

$$\|\sigma^n_j - i\bigstar\sigma^n_j\|_C \le \sum_{\ell=1}^g |\widetilde{c}^n_{j\ell}| \cdot C_\ell \cdot \eta(K_n) < M \cdot \sum_{\ell=1}^g C_\ell \cdot \eta(K_n).$$

By Theorem 4.18, for any positive constant ε_j satisfying $0 < \varepsilon_j < \|\theta_j\|_\Omega$, there exist $N_{2,j} \in \mathbb{N}$ such that

$$\|W\sigma^n_j - \theta_j\|_\Omega < \varepsilon_j,$$

for $n \ge N_{2,j}$, and therefore we obtain

$$\|\sigma^n_j\|_C = \|W\sigma^n_j\|_\Omega > \|\theta_j\|_\Omega - \varepsilon_j.$$

Hence we have

$$\begin{aligned}
\big|\rho_{i\bigstar}(\sigma^n_j) - 1\big| &= \left| \frac{\langle i\bigstar\sigma^n_j, \sigma^n_j\rangle_C}{\langle \sigma^n_j, \sigma^n_j\rangle_C} - 1 \right| \\
&\le \frac{\|\sigma^n_j - i\bigstar\sigma^n_j\|_C}{\|\sigma^n_j\|_C} \\
&< \frac{M\cdot\sum_{\ell=1}^g C_\ell}{\|\theta_j\|_\Omega - \varepsilon_j} \cdot \eta(K_n)
\end{aligned}$$

for $n \ge \max\{N_1, N_{2,j}\}$. This implies that

$$\big|\rho_{i\bigstar}(\sigma^n_j) - 1\big| = O\big(\eta(K_n)\big), \quad as\ n\to\infty$$

for $1 \le j \le g$. □

The above theorem yields two corollaries concerning the convergence rates of holomorphic 1-cochains and combinatorial period matrices.

Corollary 5.4. *Let M be a closed Riemann surface of genus $g>1$ with a canonical homology basis Σ, and let $\{K_n\}_{n\in\mathbb{N}}$ be a sequence of triangulations of M with the mesh $\eta(K_n)$ converging to zero, For arbitrary holomorphic 1-forms ω on M, we have*

$$\|W\iota_\omega^n-\omega\|_\Omega=O\big(\eta(K_n)^{1/2}\big),\quad as\ n\to\infty,$$

where $\iota_\omega^n\in\mathcal{H}C^{1,0}(K_n)$.

Proof. Any holomorphic 1-form ω can be written by

$$\omega=\sum_{j=1}^{g}\left(\int_{a_j}\omega\right)\cdot\theta_j,$$

and also we obtain

$$\iota_\omega^n=\sum_{j=1}^{g}\left(\int_{a_j}\omega\right)\cdot\sigma_j^n$$

for all $n\in\mathbb{N}$. Using Riemann's bi-linear relation of holomorphic 1-cochains, we compute

$$\begin{aligned}\langle\bigstar\sigma_j^n,\sigma_j^n\rangle_C &= \sum_{j=1}^{g}\big(\sigma_j^n(a_\ell)\overline{\sigma_j^n(b_\ell)}-\sigma_j^n(b_\ell)\overline{\sigma_j^n(a_\ell)}\big)\\ &= \overline{\pi_{jj}^{K_n}}-\pi_{jj}^{K_n}\\ &= -2i\mathrm{Im}\pi_{jj}^{K_n},\end{aligned}$$

and therefore

$$2\mathrm{Im}\pi_{jj}^{K_n}=\langle i\bigstar\sigma_j,\sigma_j\rangle_C=\rho_{i\bigstar}(\sigma_j^n)\|\sigma_j^n\|_C^2$$

By Theorem 5.3, we obtain

$$\|W\iota_\omega^n-\omega\|_\Omega\le\sum_{\ell=1}^{g}\left|\int_{a_\ell}\omega\right|\cdot\sqrt{\big(1-\rho_{i\bigstar}(\sigma_\ell^n)\big)}\cdot\|\sigma_\ell^n\|_C$$

For any positive constant ε, Theorem 4.18 gives natural numbers $N_1^1,\cdots,N_g^1$ such that

$$\|W\sigma_j^n-\theta_j\|_\Omega<\varepsilon,$$

and so

$$\|\sigma_j^n\|_C=\|W\sigma_j^n\|_\Omega<\|\theta_j\|_\Omega+\varepsilon$$

for $n\ge N_{1,j}$, where $1\le j\le g$. Also, by Theorem 5.3, for any positive constant M, there exist natural numbers $N_{2,1},\cdots,N_{2,g}$ such that

$$0<1-\rho_{i\bigstar}(\sigma_j^n)\le M\cdot\eta(K_n)$$

for $n\ge N_{2,j}$, where $1\le j\le g$. Hence we conclude

$$\|W\iota_\omega^n-\omega\|_\Omega<\sum_{\ell=1}^{g}\left|\int_{a_\ell}\omega\right|\cdot\sqrt{M}\cdot\max_{1\le j\le g}\big(\|\theta_j\|_\Omega+\varepsilon\big)\cdot\eta(K_n)^{1/2}$$

for $n\ge\max\{N_{1,1},\cdots,N_{1,g},N_{2,1},\cdots,N_{2,g}\}$. Hence the claim of this corollary follows. □

Corollary 5.5. *Let M be a closed Riemann surface of genus $g>1$ with a canonical homology basis Σ, and let $\{K_n\}_{n\in\mathbb{N}}$ be a sequence of triangulations of M with the mesh $\eta(K_n)$ converging to zero. Then Σ and $\{K_n\}_{n\in\mathbb{N}}$ yield a sequence $\{\Pi_{K_n}\}_{n\in\mathbb{N}}$ of combinatorial period matrices of M. Then we have*

$$\left|\pi_{j\ell}^{K_n}-\pi_{j\ell}\right|=O\big(\eta(K_n)^{1/2}\big),\quad as\ n\to\infty$$

for all pairs $(j,\ell)\in\{1,\cdots,g\}^2$, where $\pi_{j\ell}^{K_n}$ is the (j,ℓ)-entry of Π_{K_n} and $\pi_{j\ell}$ is the (j,ℓ)-entry of the period matrx Π of M determined by Σ.

Proof. Using Riemann's bi-linear relation, we compute

$$\begin{aligned}\langle\theta_j,\theta_\ell\rangle_\Omega &= i\langle -i\theta_j,\theta_\ell\rangle_\Omega \\ &= i\langle\star\theta_j,\theta_j\rangle_\Omega \\ &= i\sum_{k=1}^{g}\left(\int_{a_k}\theta_j\int_{b_k}\overline{\theta_\ell}-\int_{b_k}\theta_j\int_{a_k}\overline{\theta_\ell}\right) \\ &= i(\overline{\pi_{j\ell}}-\pi_{j\ell}) \\ &= 2\mathrm{Im}\pi_{j\ell}.\end{aligned}$$

By Theorem 4.12 and Cauchy-Schwarz inequality, we compute

$$\begin{aligned}\left|\pi_{j\ell}-\pi_{j\ell}^{K_n}\right| &= \left|2i\mathrm{Im}\pi_{j\ell}-\langle W\sigma_j^n,\star\theta_\ell\rangle_\Omega\right| \\ &= \left|i\langle\theta_j,\theta_j\rangle_\Omega - i\langle W\sigma_j^n,\theta_\ell\rangle_\Omega\right| \\ &= \left|\langle W\sigma_j^n-\theta_j,\theta_\ell\rangle_\Omega\right| \\ &= \|W\sigma_j^n-\theta_j\|_\Omega\cdot\|\theta_j\|_\Omega.\end{aligned}$$

Since $\iota_{\theta_j}^n=\sigma_j^n$, the claim of this theorem follows from Lemma 5.5. □

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